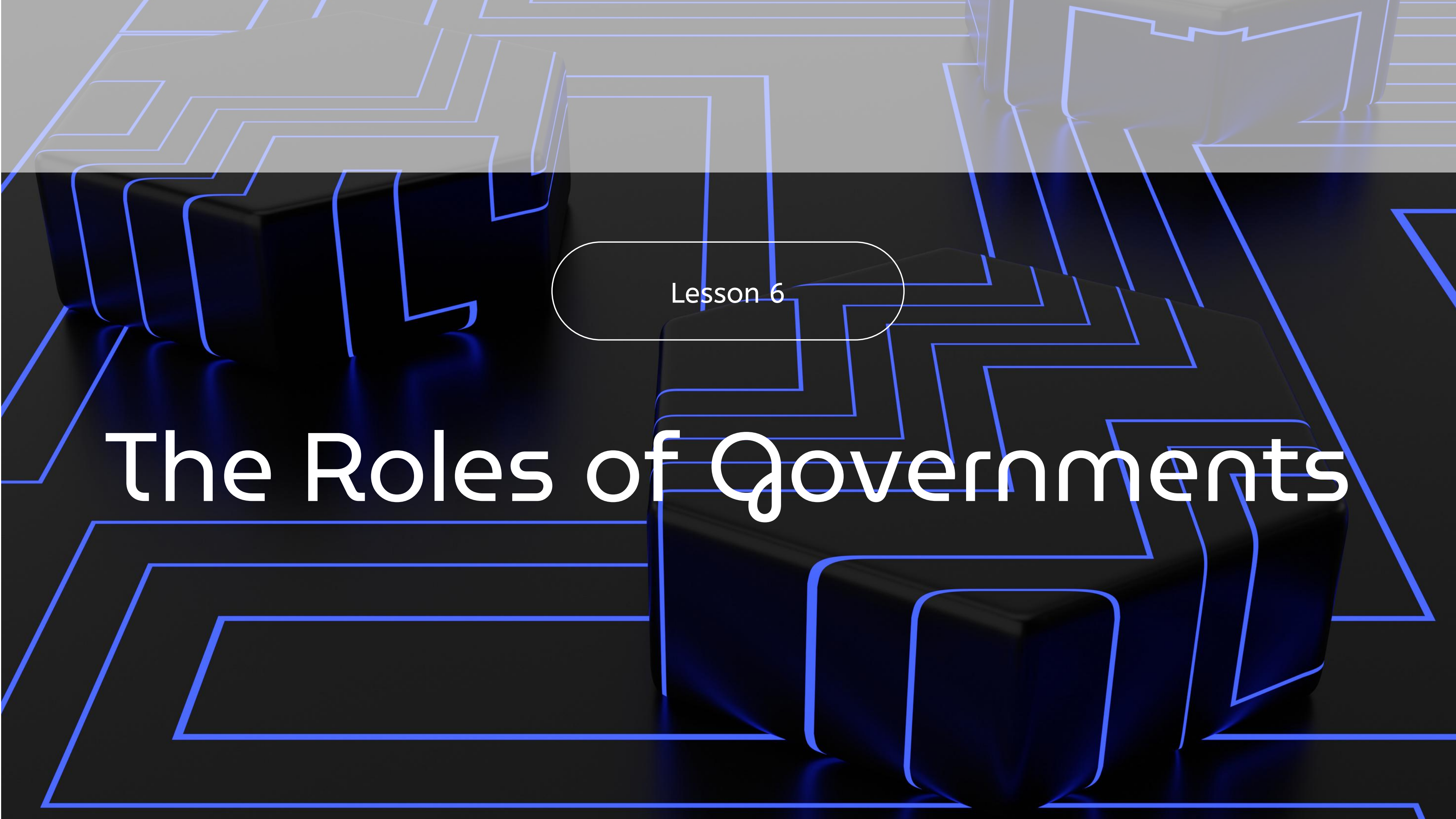


Global AI Policies and Regulatory Frameworks



全球人工智能政策 和监管框架



The background features a dark blue, almost black, surface with a complex pattern of glowing blue lines and 3D rectangular blocks. The lines form a maze-like structure, and the blocks are arranged in a way that creates a sense of depth and perspective. The overall aesthetic is modern and technological.

Lesson 6

The Roles of Governments

第6课

政府的作用

Recap of Lesson 5: AI Governance

- Foundations of AI Governance
- Structures and Institutional Roles in AI Governance
- Case Study – AI Governance in Business: The Amazon Hiring Algorithm Controversy

What we will focus on in Lesson 6: The role of governments

- Further exploring the roles of governments in AI innovation and regulation

第5课回顾： 人工智能治理

- 人工智能治理的基础
- 人工智能治理中的结构和机构角色
- 案例研究-商业中的人工智能治理： 亚马逊招聘算法争议

我们将在第六课中重点讨论的内容： 政府的作用

- 进一步探索政府在人工智能创新和监管中的作用



Agenda

1

The role of
governments in AI
innovation

2

The role of
governments in AI
regulation

3

Case Studies on
AI Regulation and
Policy
Implementation



议题

1

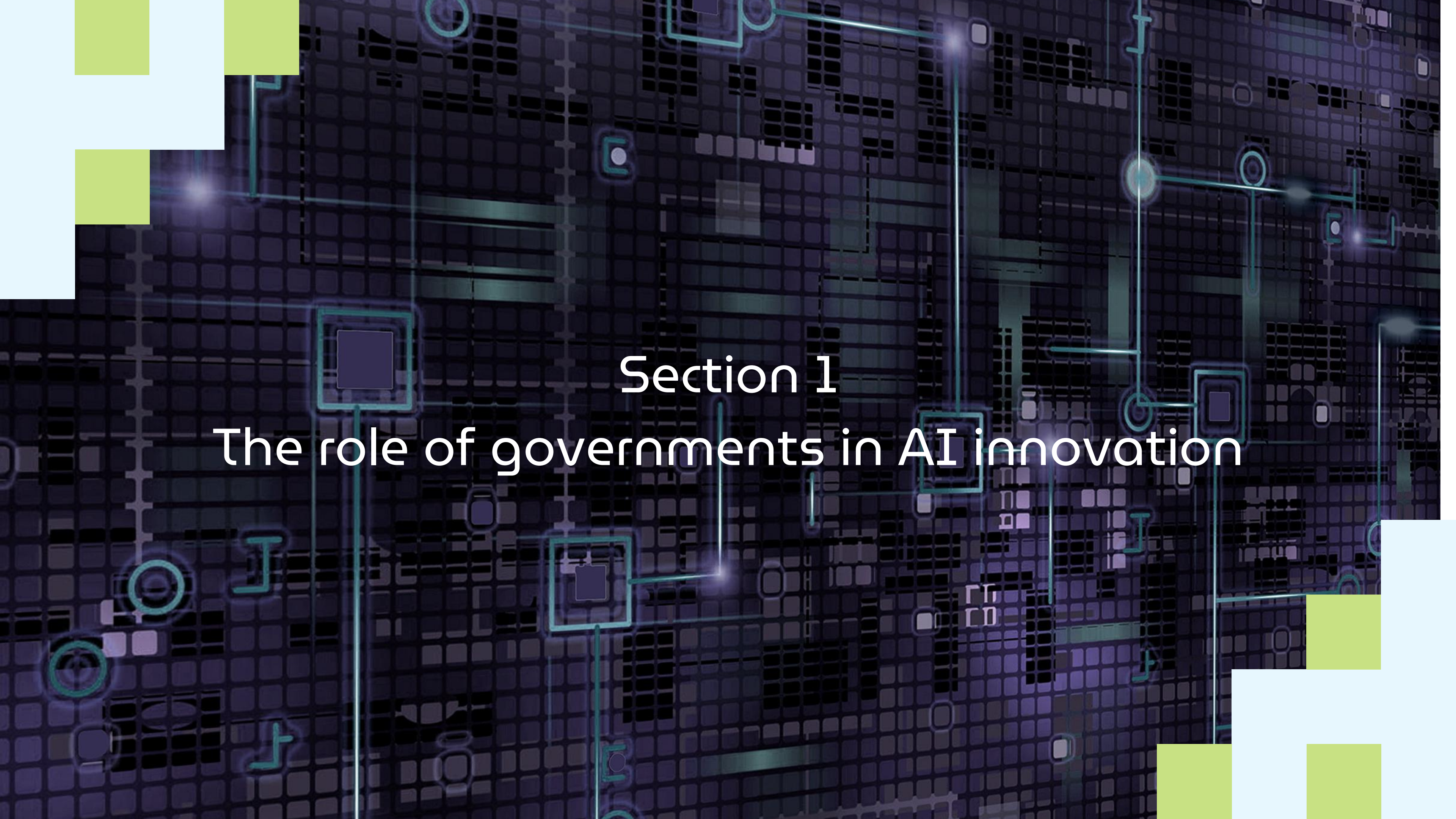
政府在人工智能创
新中的作用

2

政府在人工智能监
管中的作用

3

人工智能监管和政
策实施案例研究



Section 1

The role of governments in AI innovation



第1节 政府在人工智能创新中的作用

Summary of This Section

- Public sector as a catalyst for innovation
- national AI strategies (UK, China, EU, and Canada, USA)
- Governments enabling ecosystems
- Hidden harms of the innovation-first approach

本部分总结

- 公共部门是创新的催化剂
- 国家人工智能战略（英国、中国、欧盟和加拿大、美国）
- 政府赋能创新生态
- 创新优先方法的隐患

The Role of Governments in AI Innovation (Overview)

1

**Public sector as
enabler**

2

Balancing act

3

**Global
competition**

政府在人工智能创新中的作用（概述）

1

公共部门的赋能
作用

2

平衡之举

3

全球竞争

Public Sector as a Catalyst for Innovation

- **Direct R&D funding:** e.g., DARPA's AI research in the US.
- **AI infrastructure:** National computing platforms, open data policies.
- **Procurement as innovation tool:** Public sector adopts AI to stimulate private solutions.
- **Thomas et al. (2025):** Governments can unintentionally amplify harm if focus is solely on economic gains.



公共部门是创新的催化剂

- **直接研发资金：**例如，美国国防高级研究计划局的人工智能研究。
- **人工智能基础设施：**建设国家计算平台，推行开放数据政策。
- **采购作为创新工具：**公共部门通过应用人工智能，激励私营机构推出解决方案。
- **托马斯等人（2025）：**如果只关注经济收益，政府可能会无意中放大负面影响。



National AI Strategies (Global Examples)



国家人工智能战略（全球案例）



Governments Enabling Ecosystems

Open data and standards

Education & Talent programs

Public-private partnerships

Global collaboration

Building AI hubs & innovation
clusters

政府赋能创新生态

开放数据和标准

教育与人才计划

公私合作伙伴关系

全球协作

打造人工智能中心和创新集群

Hidden Harms of Innovation-First Approach



Social inequities

Uneven access to AI benefits, digital divide



Labour displacement

Automation risk without social safety nets



Environmental cost

Energy consumption from AI model training



Surveillance & Privacy Risks

The adoption of invasive technologies e.g. facial recognition, predictive policing

创新优先方法的隐患

社会不平等

获得人工智能好处的机会不均衡，数字鸿沟

劳动力替代

没有社会安全网的自动化风险

环境成本

AI模型训练的能耗

监控和隐私风险

采用侵入性技术，如面部识别、预测性警务

Key Takeaways

- Governments shape what AI gets built, by whom, and for whose benefit through investment, infrastructure, and standard- setting—long before regulation bites.
- Innovation without governance attention risks hidden harms and social backlash that can derail adoption.

Strategic emphases differ:

- UK = pro- innovation + standards;
- EU = coordinated investment + trust;
- U.S. = R&D + industrial policy;
- China = state- guided scale + governance;
- Canada = talent + responsible commercialization.

关键点

- 政府通过投资、基础设施建设与标准制定，早在监管生效之前，就已深刻决定了人工智能的开发方向、开发主体及受益对象。
- 若缺乏治理关注，创新可能引发隐性危害与社会反弹，最终反而会阻碍其推广应用。

战略重点不同：

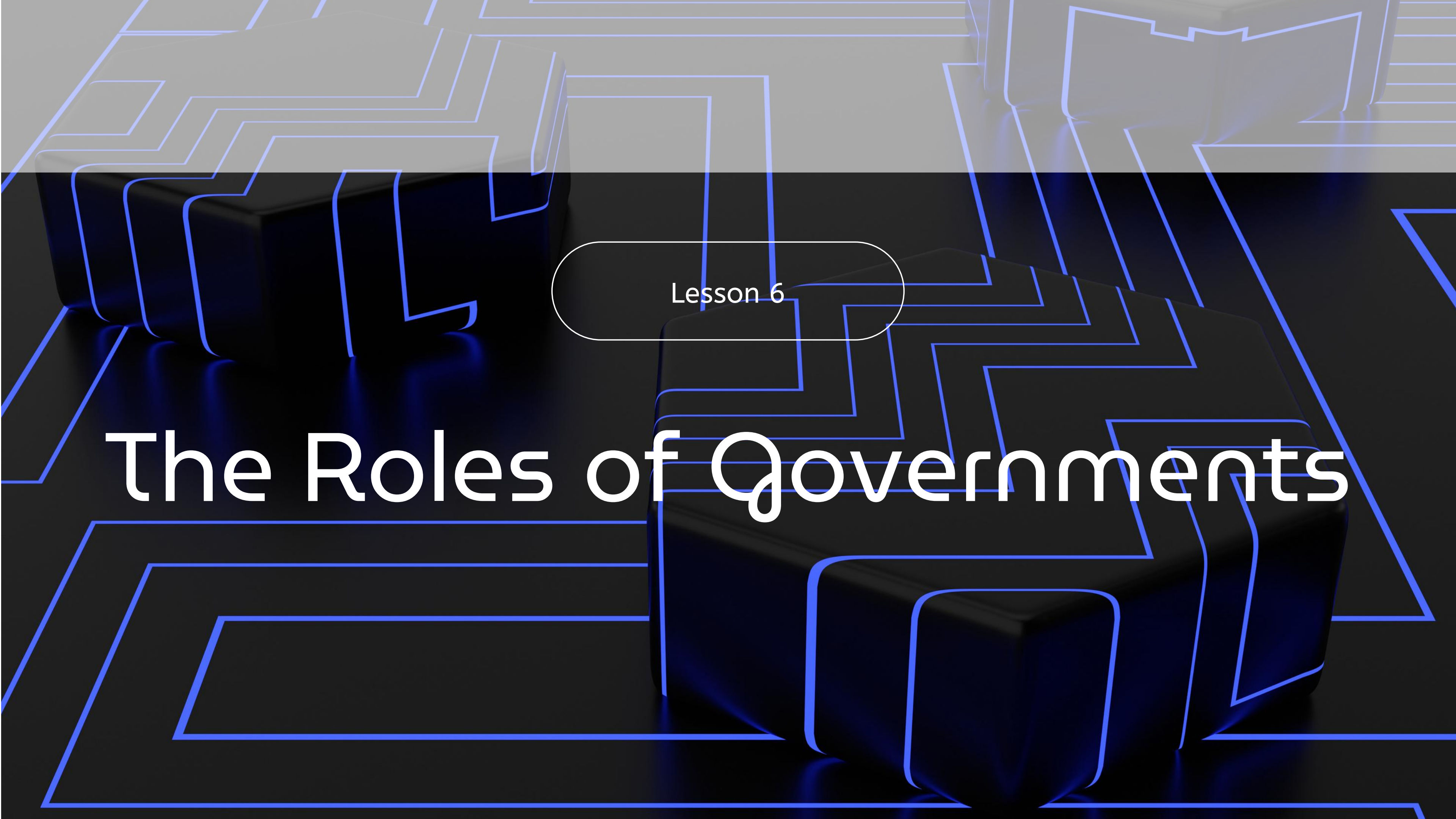
- 英国=支持创新+建立标准；
- 欧盟=协调投资+信任框架；
- 美国=研发投入+产业政策；
- 中国=国家引导下的规模化发展+治理；
- 加拿大=人才+负责的商业化。

THANKS !!!



谢谢!!!



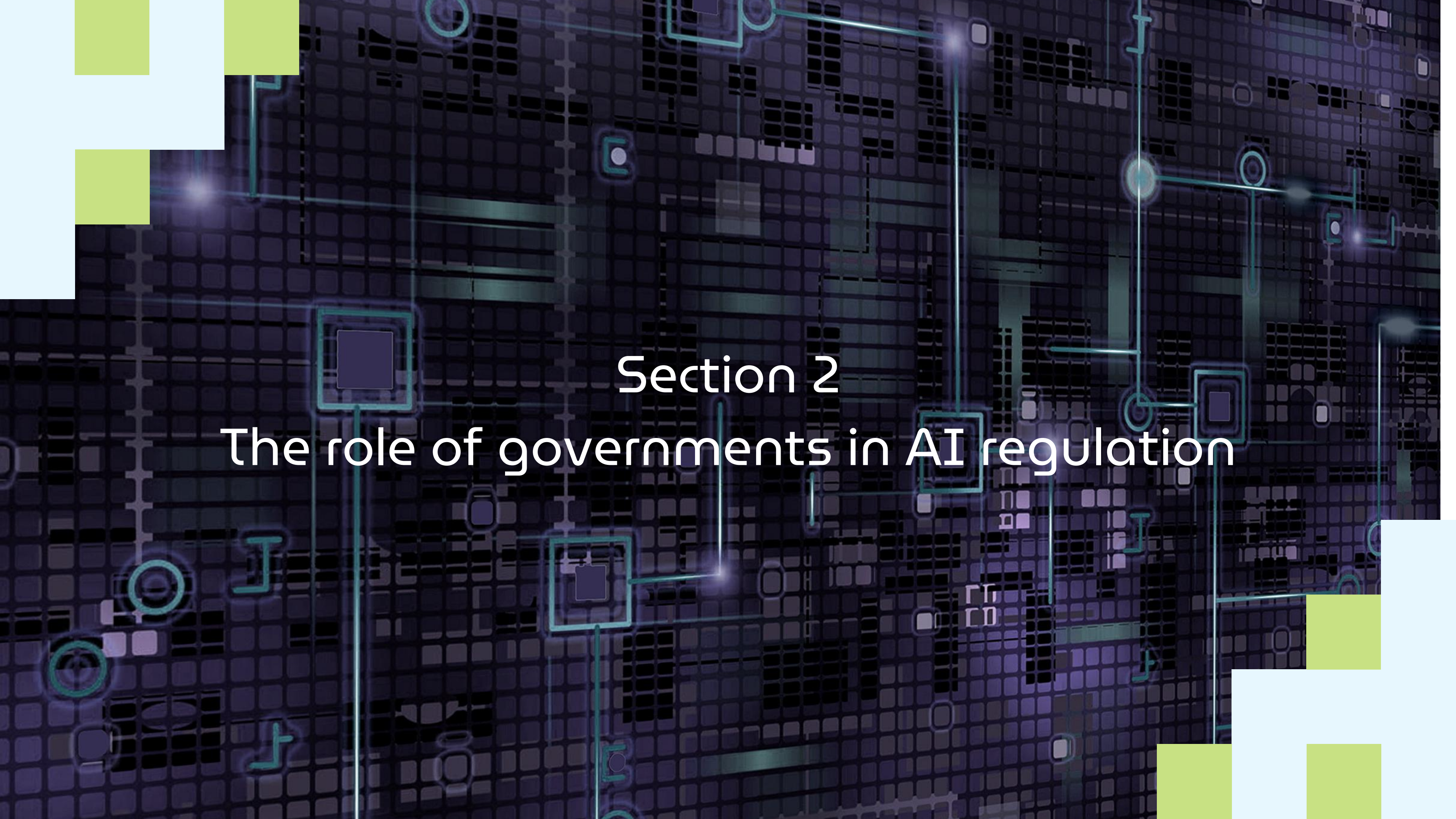
The background features a complex, abstract design of blue lines and geometric shapes, resembling a stylized cityscape or a network of paths. The lines are of varying thickness and create a sense of depth and movement. The overall color palette is dark blue and black, with the blue lines providing a strong contrast.

Lesson 6

The Roles of Governments

第6课

政府的作用



Section 2

The role of governments in AI regulation



第2节

政府在人工智能监管中的作用

Summary of This Section

- Comparative regulatory approaches (EU, US, China)
- Core governance tensions
- Ethical outcomes in regulatory practice
- Global influence of EU law

本部分小节

- 比较监管方法（欧盟、美国、中国）
- 治理核心矛盾
- 监管实践中的伦理结果
- 欧盟法律的全球影响力

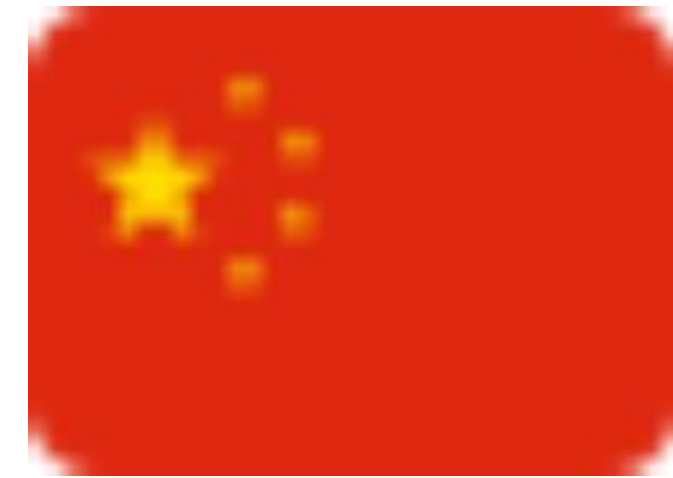
Comparative Regulatory Approaches



- AI Act (risk-based model), focus on fundamental rights, transparency.
- Synergy with GDPR (e.g., Article 22 – automated decision rights).



- Fragmented, sector-specific (e.g., FTC guidance, NIST framework).
- Innovation-first approach with voluntary standards.



- Proactive content moderation, algorithmic recommendation rules.
- AI governance tied to social stability and state control.

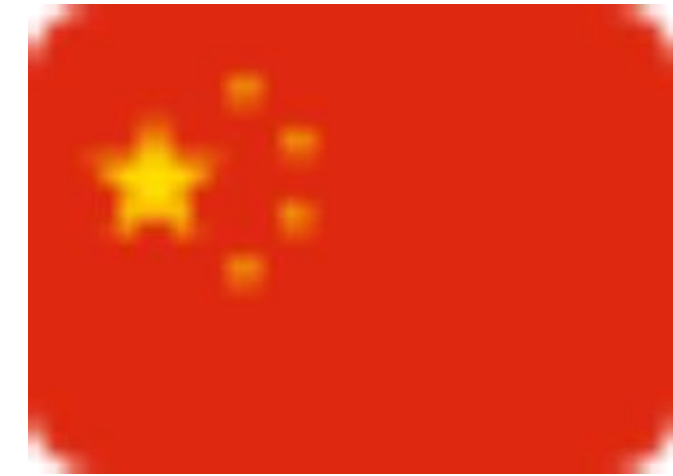
比较监管方法



- 《人工智能法案》（基于风险的模型），聚焦基本权利与透明度。
- 与《通用数据保护条例》的协同效应（例如第 22 条 —— 自动化决策权相关规定）。



- 碎片化且分行业监管（例如美国联邦贸易委员会（FTC）相关指引、美国国家标准与技术研究院（NIST）《人工智能风险管理框架》）。
- 采用自愿标准的创新优先方法。



- 主动开展内容治理，规范算法推荐行为。
- 人工智能治理与社会稳定和国家管控息息相关。

Core Governance Tensions

**Ethics
vs.
Economic
Growth**

**Sovereignty
vs.
Harmonisation**

**Short- Term
Risks vs.
Long- Term
Accountability**

**National
Security vs.
Open Innovation**

治理核心矛盾

伦理
vs.
经济增长

主权自主
vs.
协调统一

短期风险
vs.
长期问责

国家安全
vs.
开放式创新

Ethical Outcomes in Regulatory Practice

Laws aim to:

- Ensure fairness, accountability, and explainability.
- Avoid “ethics washing” by requiring impact assessments and human oversight.



监管实践中的伦理结果

法律旨在：

- 确保公平性、问责性和可解释性。
- 通过要求开展影响评估并实施人工监督，避免“伦理粉饰”。



Global Influence of EU Law

- **The Brussels Effect:** EU laws set global de facto standards due to market size.
- Countries and companies adopt EU standards (e.g., GDPR influence).
- Emerging economies look to EU for guidance on risk classification and consumer rights.



欧盟法律的全球影响力

- **布鲁塞尔效应：** 由于市场规模，欧盟法律设定了全球事实上的标准。
- 各国及企业纷纷采用欧盟标准（例如受《通用数据保护条例》的影响）。
- 新兴经济体期待欧盟在风险分类和消费者权益方面提供指导。



Global Influence of EU Law



欧盟法律的全球影响力

Key Takeaways

- EU = Binding, risk- tiered, rights- driven; U.S. = Fragmented + executive + sector enforcement; China = Fast iterative domain rules + state oversight.
- Regulatory design encodes ethical priorities; enforcement + assurance determine real outcomes.
- The EU's AI Act may project a new Brussels Effect, influencing draft laws and industry practice globally—watch Canada's AIDA and UK alignment efforts

关键点

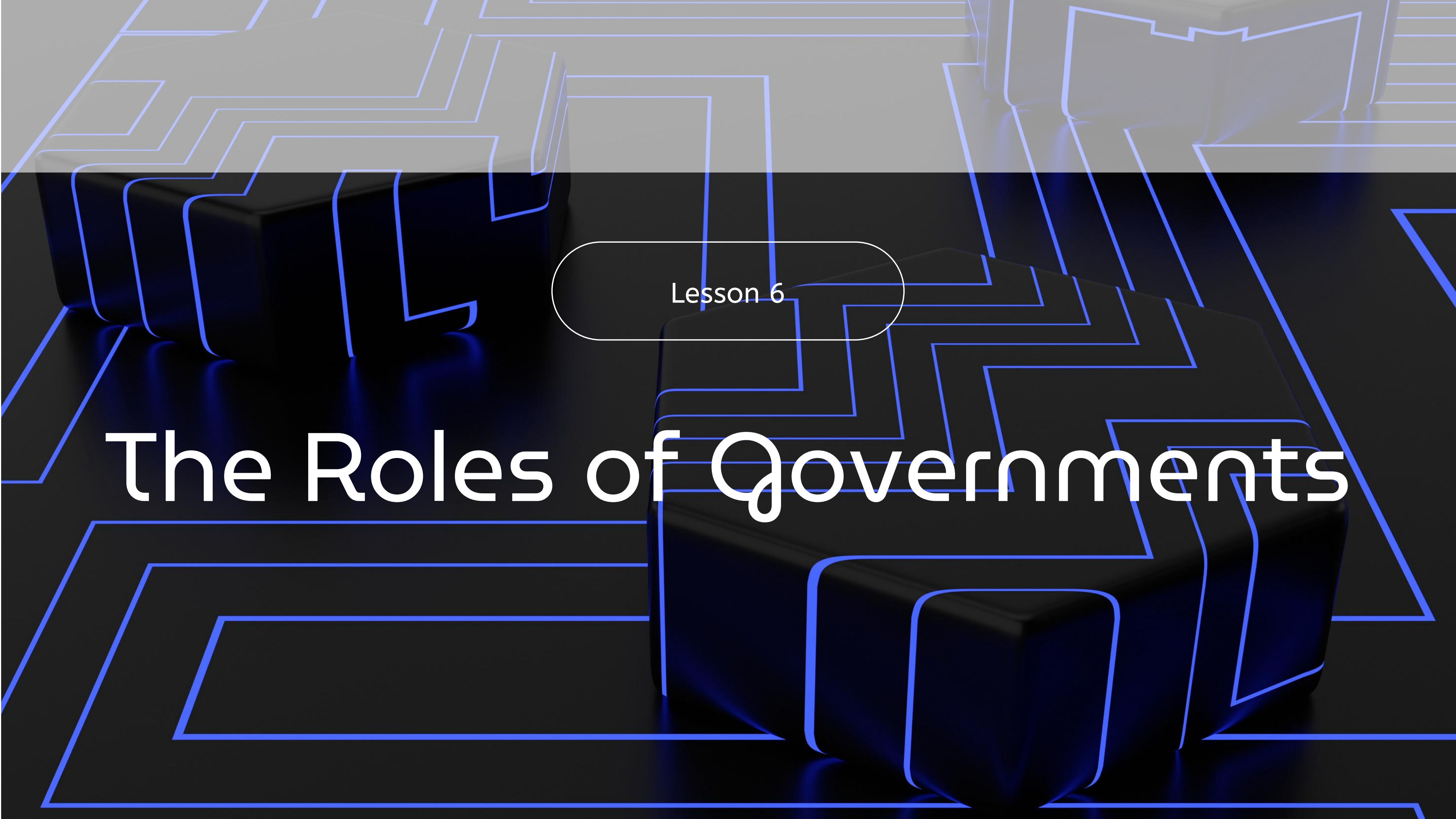
- 欧盟=具有约束力、风险分级、权利驱动；美国=碎片化+行政+部门执法；中国=快速迭代的领域规则+国家监管。
- 监管设计体现了伦理优先序；执行力与保障措施共同决定了实际成效。
- 欧盟的人工智能法案可能会产生新的布鲁塞尔效应，影响全球法律草案和行业实践——关注加拿大《人工智能与数据法》和英国的协调努力

THANKS !!!



谢谢!!!



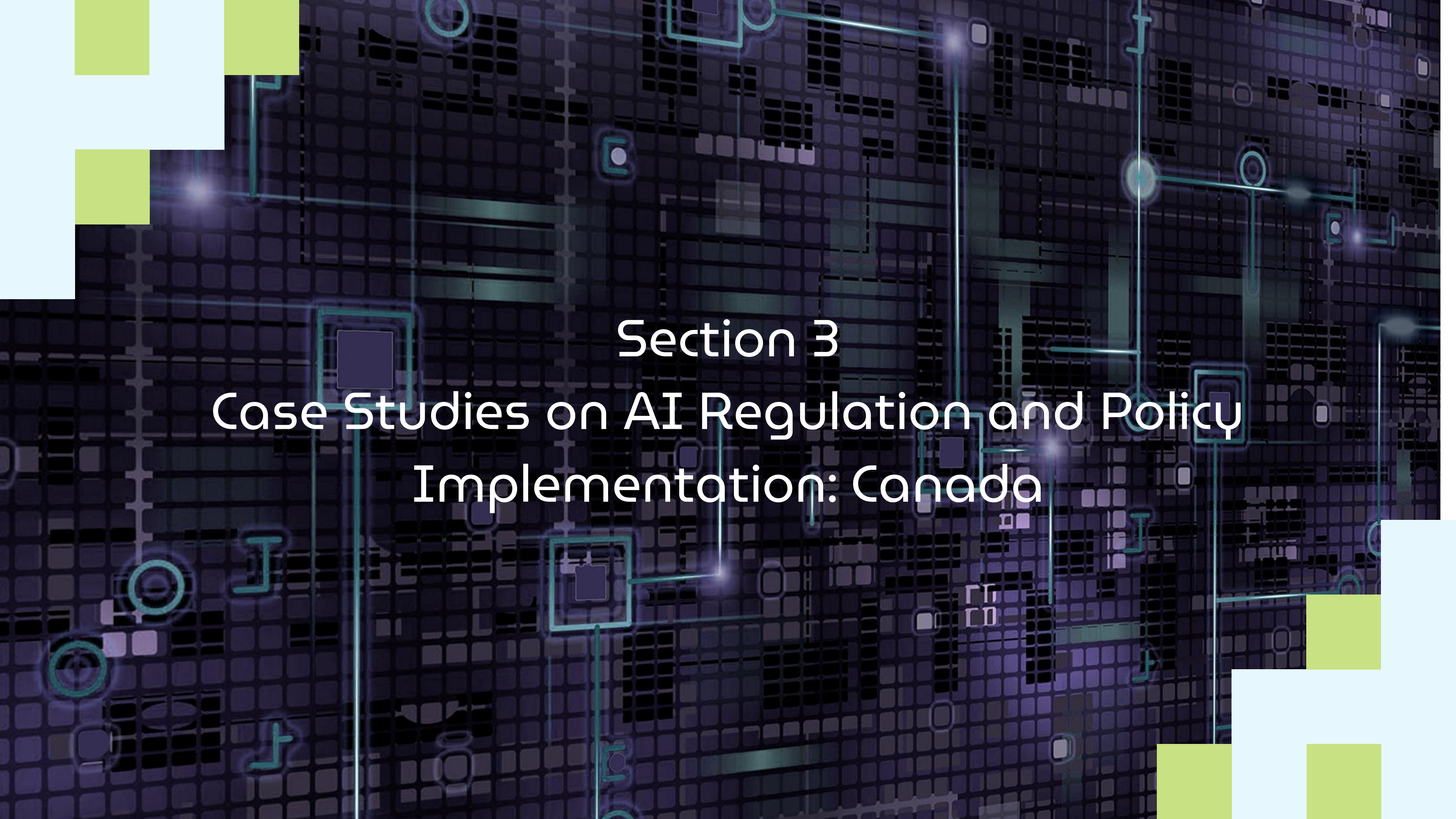
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Lesson 6

The Roles of Governments

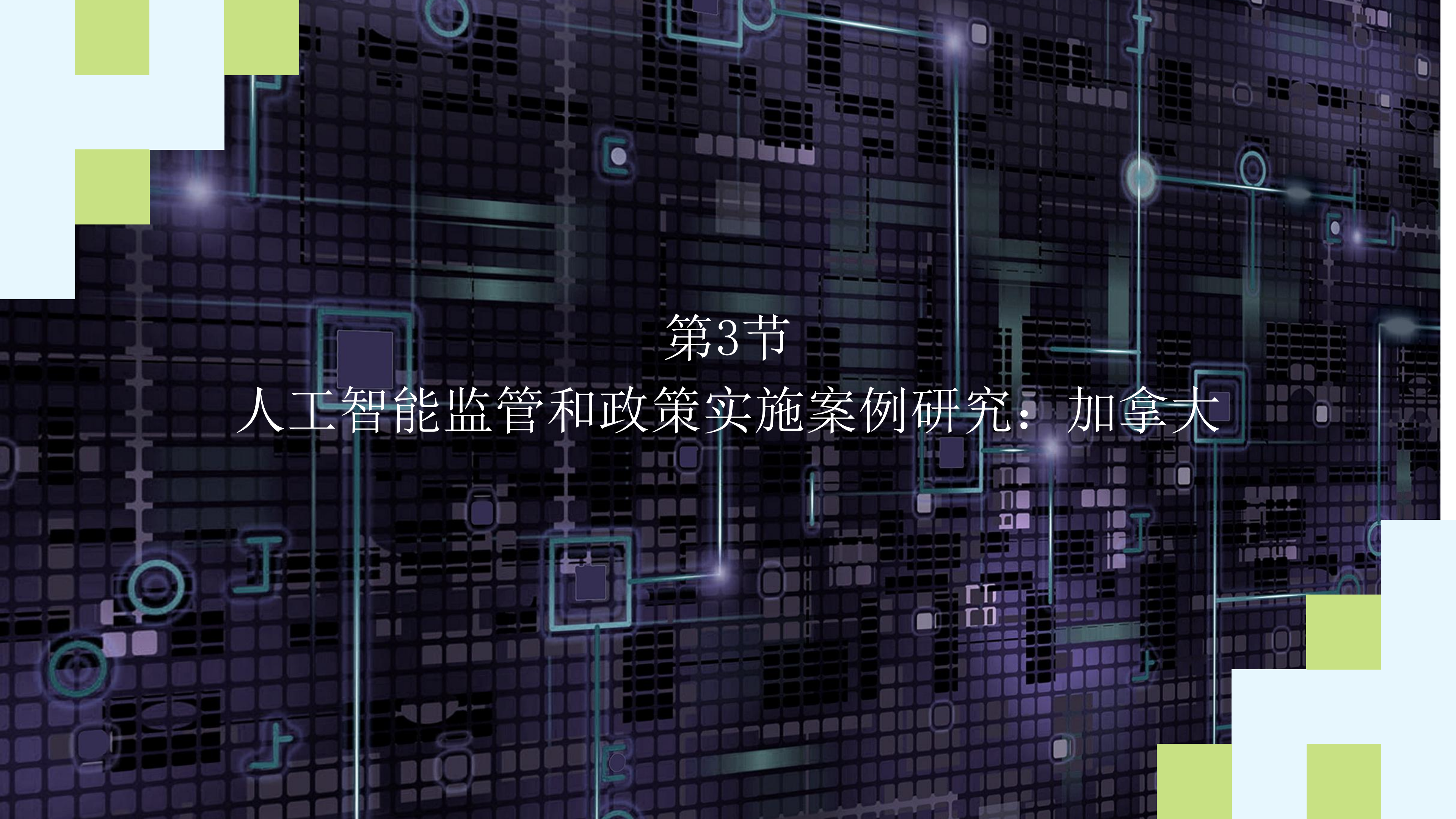
第6课

政府的作用



Section 3

Case Studies on AI Regulation and Policy Implementation: Canada



第3节

人工智能监管和政策实施案例研究：加拿大

Summary of This Section

- Overview of Canada's AI regulatory landscape (AIDA)
- Principles behind AIDA
- Implementation challenges
- Comparison with other global frameworks

本部分小节

- 加拿大人工智能监管格局概述（基于《人工智能与数据法》）
- 《人工智能与数据法》背后的原则
- 实施挑战
- 与其他全球框架的比较

AIDA Timeline



《人工智能与数据法》 时间表



Why AIDA?

- Existing sectoral statutes left gaps for cross-cutting AI risks;
- Canadians are both concerned about AI harms and receptive to government oversight;
- The stated aim: build trust, support responsible innovation, and ensure international interoperability.



为何推出《人工智能与数据法》？

- 现有的部门法规为跨领域人工智能风险留下了空白；
- 加拿大人既担心人工智能的危害，又接受政府的监督；
- 既定目标：建立信任，支持负责任的创新，并确保国际互操作性。



Core Design & Principles Behind AIDA

**Risk- Based,
Harm- Mitigatio
n Focus**

**Lifecycle
Accountability**

**Transparency &
Record- Keepin
g**

**Bias & Harm
Mitigation Duties**

**Agile,
Regulation- Lite
Statute + Detailed
Regulations Later**

**Interoperability
with Global
Regimes**

《人工智能与数据法》的核心设计和原则

以风险为基础，以
减轻危害为重点

生命周期责任

透明度和记录保存

偏见和损害缓解义
务

敏捷化、轻量型法
规+后续出台详细规
章

与全球制度的互操
作性

Implementation Challenges

Legislative Process Fatigue & Political Timing

Perceived Vagueness & Over- Delegation

Definition & Scope Debates

Oversight Architecture & Independence

Alignment with Existing Privacy & Sectoral
Rules

Stakeholder Inclusion & Equity Concerns

SME Readiness & Standards Gap

实施挑战

立法进程推进乏力与政治时机制约

感知模糊和过度授权

定义和范围争论

监督架构和独立性

与现有隐私和行业规则保持一致

利益相关者包容和公平关切

中小企业准备度与标准差距

Global Comparison



- EU AI Act: Risk tiers, strong enforcement, extraterritorial reach
- AIDA: Narrower scope, focused on commerce-related high-impact systems
- EU: AI Office + national market surveillance; AIDA: Minister & Commissioner under ISED

- US: No federal AI law; executive orders & NIST RMF
- China: Iterative AI rules (algorithm recommendations, deep synthesis)
- AIDA: Seeks interoperability with both US & EU while preserving Canadian priorities

全球比较



- 欧盟《人工智能法案》：基于风险分级、执法严厉并具有域外效力。
- 加拿大《人工智能与数据法》：范围较窄，专注于商业相关的高影响系统。
- 欧盟：设立“人工智能办公室”并协同各国市场监管机构；加拿大则由创新、科学和工业部（ISED）下属的部长及专员负责。

- 美国：暂无联邦层面人工智能专项法律，主要依靠行政令及美国国家标准与技术研究院《人工智能风险管理框架》（NIST RMF）进行规范。
- 中国：AI规则迭代（算法推荐、深度合成）。
- 《人工智能与数据法》：寻求与美国和欧盟的互操作性，同时坚守加拿大的核心优先事项。

Practical Takeaways for Organizations Operating in / from Canada

- **Don't wait for legislation**
- **Leverage Standards Infrastructure**
- **Integrate Model Risk & Human Rights Lens**
- **Scenario Plan for Multiple Futures**



面向在加拿大运营或源自加拿大的组织的实操要点

- 勿坐等立法落地
- 利用标准基础设施
- 整合模型风险与人权视角
- 为多种未来场景制定预案



Recap: The role of governments

- **The role of governments in AI innovation:**
 - a. public sector as a catalyst for innovation
 - b. national AI strategies (UK, China, EU, and Canada, USA)
 - c. governments enabling ecosystems
 - d. hidden harms of innovation-first approach
- **The role of governments in AI regulation**
 - a. comparative regulatory approaches (EU, US, China)
 - b. core governance tensions
 - c. ethical outcomes in regulatory practice
 - d. global influence of EU law
- **Case Study**

回顾：政府的作用

- 政府在人工智能创新中的作用：
 - a. 公共部门作为创新的催化剂
 - b. 国家人工智能战略（英国、中国、欧盟和加拿大、美国）
 - c. 政府赋能创新生态
 - d. 创新优先方法的隐患
- 政府在人工智能监管中的作用
 - a. 比较监管方法（欧盟、美国、中国）
 - b. 治理中的核心矛盾点
 - c. 监管实践中的伦理结果
 - d. 欧盟法律的全球影响力
- 案例研究

Course Recap

- **Risk + Rights + Context:** Technical risk must be braided with rights protections and domain context.
- **Process Legitimacy Matters:** Inclusion, transparency, and trust in institutions shape regulatory durability as much as statutory text.
- **Global Interdependence:** Even domestic rules are drafted in the shadow of global markets — the ‘Brussels effect,’ standards convergence, and supply- chain compliance all matter.

课程回顾

- **风险+权利+场景：** 技术风险必须与权利保护、具体领域场景深度融合。
- **程序合法性至关重要：** 包容性、透明度及对机构的信任，与法定文本同等影响监管的持久性。
- **全球相互依存：** 即使是国内规则，也会受全球市场环境的影响而制定——“布鲁塞尔效应”、标准趋同和供应链合规性都很重要。

THANKS !!!



谢谢!!!

